

Network Services Configuration -- Guided Notes

Unit 2.5.1 | CompTIA Network+ N10-009 Obj. 2.5 | N10-008 2.5

Name: _____ Date: _____ Period: _____

Complete each question using the lesson reading. Write in the blank or answer in the space provided.

Question 1

A DHCP server automatically assigns _____, _____, _____, and _____ to clients on a network. Without DHCP, these values would need to be configured _____ on every device.

Question 2

The four steps of the DHCP process in order are: _____, _____, _____, _____. The memory aid for this sequence is _____.

Question 3

A DHCP scope defines the _____ of addresses available for assignment. An exclusion range removes specific addresses from the pool (typically used for _____). A DHCP reservation assigns the _____ IP address to a device based on its MAC address.

Question 4

When a client on one subnet needs to obtain a DHCP address from a server on a different subnet, a _____ agent must be configured on the router interface. This agent forwards DHCP _____ packets across subnet boundaries since these packets cannot normally cross a router.

Question 5

DNS resolves _____ names (like www.example.com) to _____ addresses. A DNS _____ record maps a hostname to an IPv4 address. A _____ record maps a hostname to an IPv6 address. A _____ record maps an IP address back to a hostname.

Question 6

NTP (Network Time Protocol) synchronizes _____ across network devices. Accurate time is critical for _____, _____, and _____. NTP uses a hierarchy of servers called _____ levels, with

atomic clocks at stratum _____.

Question 7

Explain the difference between a DHCP lease and a DHCP reservation. When would you use each, and what happens when a DHCP lease expires if the client is still connected?

Question 8 -- Real World Application

REAL WORLD: Users on a network segment report that some devices receive IP addresses in the expected range (192.168.10.x) but others receive addresses in the 169.254.x.x range. Explain what 169.254.x.x addresses indicate, list three possible causes, and describe the first diagnostic command you would run.

Download the answer key from the class server:

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wget http://[SERVER]/resources/network-engineer/2-5-1-network-services-configuration/2-5-1-network-services-configuration-answer-key.pdf
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